

# Ayush Mondal

ayushmondal139@gmail.com — +91 9051217331 — [github.com/ayushh9999](https://github.com/ayushh9999) — [linkedin.com/in/ayushmondal139](https://linkedin.com/in/ayushmondal139)

## Education

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| <b>RCC Institute of Information Technology (RCCIIT)</b><br><i>B.Tech in Electronics and Communication Engineering (7.9)</i> | <i>2022 – present</i><br><i>Kolkata, WB</i> |
| <b>Purwanchal Vidya Mandir</b><br><i>Secondary (90%) and Higher Secondary Education (74%)</i>                               | <i>2020–2022</i><br><i>Kolkata, WB</i>      |

## Work Experience

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### Codec Technologies

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| <i>Machine Learning Engineer Intern</i>                                                                                                                                                                                                                                                                | <i>Kolkata, WB</i> |
| <ul style="list-style-type: none"><li>Developed an end-to-end Customer Churn Prediction model using Python and Scikit-learn, achieving 87% accuracy through data preprocessing, feature engineering, model training, and evaluation. Prepared the trained model for deployment using Joblib.</li></ul> |                    |

### Consistent Services

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| <i>Software Engineer Intern</i>                                                                                                                                                                                                                                                                                                                                                          | <i>Kolkata, WB</i> |
| <ul style="list-style-type: none"><li>Developing an interactive 3D WAP-7 locomotive visualization system using Unity/Unreal Engine, simulating traction, braking, auxiliary and pneumatic systems with real-time animations, selectable components and exploded views, while integrating bilingual audio-visual training modules and adhering to RDSO/CLW technical standards.</li></ul> |                    |

### Allotrope Systems Private Limited

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| <i>Backend Engineer Intern</i>                                                                                                                                                                                                                                                                                                                                     | <i>Kolkata, WB</i> |
| <ul style="list-style-type: none"><li>Developed a Hospital Management System backend using Node.js, Express.js, and MongoDB, implementing REST APIs and Mongoose schemas for core workflows (appointments, billing, beds, discharge), while applying MVC architecture, async/await, and centralized error handling for scalable and maintainable design.</li></ul> |                    |

## Projects

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| <b>Facial Recognition System (Python, OpenCV, CVzone)</b>                                                                                                                                                                                                                                                         | GitHub |
| <ul style="list-style-type: none"><li>Developed a real-time facial-recognition attendance system using Python, OpenCV, and face recognition, leveraging pre-trained embedding models for identity matching and engineering a reliable check-in workflow for automated attendance logging and reporting.</li></ul> |        |

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| <b>Portfolio Assistant (Python, LangChain, FAISS, MistralAI)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                   | GitHub   Deployed Link   Demo Video |
| <ul style="list-style-type: none"><li>Developed a RAG-powered Portfolio Assistant using Python, LangChain, Mistral AI, FAISS, and Streamlit that enables recruiters to ask natural language questions about a candidate's CV. Implemented document chunking, embedding generation with Sentence Transformers, semantic search using FAISS, and context-aware response generation through Mistral AI. Built both terminal and web-based interfaces for interactive CV querying and information retrieval.</li></ul> |                                     |

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| <b>AWS Data Pipeline (AWS S3, Lambda, Glue, EventBridge, SNS, CloudWatch, Python)</b>                                                                                                                                                                                                                                                                             | GitHub |
| <ul style="list-style-type: none"><li>Developed a Python-based serverless ETL pipeline using AWS Glue and AWS Lambda to ingest, clean, and transform raw CSV data into analytics-ready outputs, implemented modular workflow components for maintainability and faster iteration, and improved data quality while reducing manual preprocessing effort.</li></ul> |        |

## Technical Skills

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| <b>Languages:</b> Python, SQL, JavaScript, Java                                                                                                                                     |  |
| <b>Frameworks/Libraries:</b> Scikit-Learn, Pandas, Computer Vision, OpenCV, YOLO, Matplotlib, NLP, LangChain, RAG Pipelines, AI Agents, TensorFlow, Express.js, LLM, FastAPI, GenAI |  |
| <b>Databases:</b> MongoDB, VectorDB, ChromaDB, PostgreSQL, MySQL                                                                                                                    |  |
| <b>Tools:</b> Git, GitHub, AWS, Postman, Ollama, Mistral, Gemini API                                                                                                                |  |
| <b>Certifications &amp; Achievements</b>                                                                                                                                            |  |

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**AMCAT (2025):** Automata 93/100, Advanced Quant 84/100, Programming 79/100 — AWS Cloud Practitioner (GFG)  
— LangChain: Foundations (LangChain Academy) — Intro to Modern AI (Cisco) — Intro to Python (Nasscom)